



TECHNICAL DATA SHEET

ProLine PPS-CF Filament

Product Overview

PPS-CF is a high-performance 3D-printing filament based on a modified polyphenylene sulfide (PPS) compound with 10% carbon-fiber reinforcement. It combines high temperature resistance, very low moisture absorption, good chemical resistance and inherent flame retardancy. The material can be processed on suitable 3D printers even without a heated build chamber.

Printed parts show high rigidity and strength. The heat deflection temperature can reach up to 245 °C, with a continuous use temperature of around 220 °C. Due to the low moisture uptake, dimensional stability and electrical properties are comparatively less influenced by humidity and temperature.

Key Features

- Rigid, high-strength parts
- Performs reliably at very high temperatures
- Inherently flame-retardant
- Low moisture uptake with good chemical resistance

Material Properties

Physical Properties			
Property	Test Method	Unit	Value
Density	ISO 1183	g/cm ³	1.30-1.32
MFR (316°C / 5 kg)	ISO 1133	g/10 min	50-60
Moisture Absorption (23°C / 24 h)	ISO 62	%	< 0.1
Flame retardancy	UL 94	1/16"	V-0
Mechanical Properties			
Tensile strength (X-Y)	ISO 527	MPa	95-105
Elongation at break (X-Y)	ISO 527	%	4-6
Flexural modulus (X-Y)	ISO 527	MPa	8000-9000

Flexural strength (X-Y)	ISO 178	MPa	125-135
Flexural strength (Z)	ISO 178	MPa	50-52
Flexural modulus (X-Y)	ISO 178	MPa	6000-6500
Flexural modulus (Z)	ISO 178	MPa	2200-2260
Impact strength (X-Y)	ISO 180	kJ/m ²	7-8
Notched impact strength (Z)	ISO 180	kJ/m ²	2.5-3
Thermodynamic Properties			
HDT @ 0.455 MPa (66 psi)	ISO 75	°C	245
Continuous Use Temperature	IEC 60216	°C	220
Electrical characteristics			
Surface resistance	IEC 60093	Ω	≤ 10 ⁵

Recommended Printing Parameters

Parameter	Setting
Print Temperature	310-350°C
Bed Temperature	100-120°C
Bed Materials	Tempered glass, BuildTak, Carbon fiber board
Nozzle Diameter	Ø 0.4 / 0.6 mm
Nozzle and gear material	High strength steel
Model Cooling Fan	0-30%
Printing Environmental Temperature	Room temperature to ~80°C

Drying & Handling Notes

- Pre-dry: 120°C, minimum 8 h (hot-air oven) to reduce moisture-related printing issues and improve surface/part quality
- Post-dry: 80-100°C, 1-3 h to increase part strength

Disclaimer

Product information and test results in this document were obtained under controlled laboratory conditions. Real-world performance may vary depending on processing, environment, and application. The customer is responsible for determining product suitability for the intended use and for ensuring compliance with all applicable laws and regulations. The Company accepts no liability for the use of this information and provides no warranties, including any implied warranties of merchantability or fitness for a particular purpose, to the extent permitted by law.

Test Specimens

Test sample dimensions and printing direction

Figure 1. Tensile testing specimen; ASTM D638 (ISO 527, GB/T 1040)

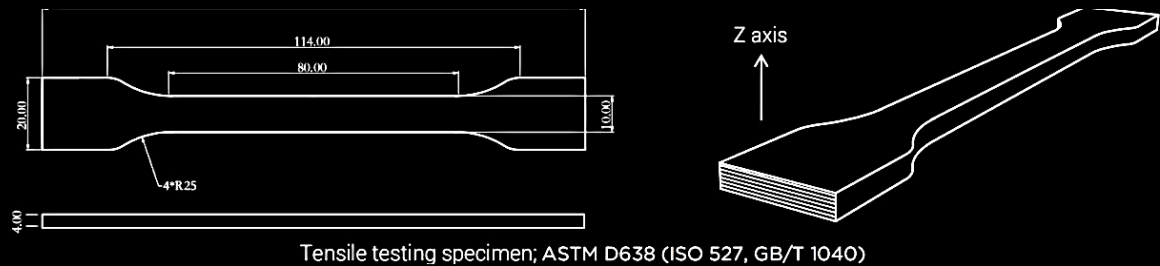


Figure 2. Flexural and impact testing specimens; ASTM D790 (ISO 178, GB/T 9341) and ASTM D256 (ISO 179, GB/T 1043)

